

MEL EDGELL - ENGINEERING PHYSICS

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PROFILE

Second-year UBC Engineering Physics student with experience designing and implementing control systems, embedded electronics, and electromechanical prototypes through academic projects. Proficient in CAD, MATLAB, and technical problem-solving, with a demonstrated ability to adapt quickly to challenging situations, learn new technologies quickly, and contribute to project success. Proven ability to communicate complex ideas clearly and collaborate seamlessly within multidisciplinary teams to bring projects from concept to reality.

EDUCATION

Engineering Physics, BSc. (in-progress)

UBC, Vancouver

Key Courses: Linear Circuits; Principles of Software Construction; Experimental Techniques

SKILLS

Mechanical: CAD (SolidWorks), Sheet Metal Fabrication

Software: C, MATLAB, Google Sheets & Excel

Hardware: Arduino, Circuit Analysis, Oscilloscopes

TECHNICAL EXPERIENCE AND PROJECTS

Cardboard Chair

- Designed a structurally efficient cutting pattern for a children's cardboard chair to maximize the load bearing capacity while minimizing the amount of material used
- Added personalized elements and children's games built into the chair design to increase interaction and engagement
- Created a technical video pitch to market the chair's features and raise awareness about the dangers of displacement due to natural disasters

Cycling Lane City Development Project

- Researched and developed a robust cycle lane development and implementation plan to improve sustainability in Vancouver
- Created a high-quality, informative presentation to share the benefits and challenges associated with a particular cycle lane plan

Assistive Technology CAD Prototype

- Developed a SolidWorks prototype of an assistive device to allow someone with limited physical mobility to dispense toothpaste from the
- Created and presented an information sheet and 1-minute elevator pitch highlighting the device's mechanical features and its importance

Automated Claw

- Successfully developed a C program that took external input of the distance to the ground and instructed Arduino components operating the claw mechanism to capture and release objects of various shapes and sizes
- Built a small aluminum-cardboard prototype that was durable and maintained enough strength to carry objects up to 800g

Mining Development Proposition

- Investigated three mining site models and various mining techniques to develop a Vancouver mine long-term development plan
- Performed streamlined life cycle assessments for all possible mine plans to determine the environmental impacts of each stage of development
- Prepared the mine site analysis portion of the recommendation report summarizing our research and chosen solution to consolidate our findings in a digestible format

Rainwater Harvester System Model

- Created a robust spreadsheet model of a rainwater harvester system that took in design inputs as variables and outputted the satisfaction values of stakeholder needs
- Developed a detailed E-poster that outlined my team's decision-making process and justified our chosen solution

Workday to Google Calendar Google Gemini-Trained Python Code

- Trained Google Gemini to develop a python code that converted Workday classes to a calendar file that can be uploaded to Google Calendar
- Effectively prompted Gemini to make modifications to the python code to improve user interface and made manual improvements

LEADERSHIP EXPERIENCE

Section Head and Summer Camp Counselor

Jul. 2024 – Aug. 2026

- Managed a team of 10 counsellors to support and care for 48 campers over the course of the summer
- Gave feedback to staff to help them improve, grow, and shape them as counsellors so they were well equipped to provide the best experience for their campers
- Communicated effectively and frequently with my staff, coworkers, and supervisors to ensure efficient problem solving